Differential Geometry

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Lecture 1

1 Manifolds

The concept of a manifold is intended to generalise curves and surfaces in $\mathbb{R}^3$.

Definition 1.1 (smooth structure)
A *smooth structure* on a topological space M is a collection of *charts* (an *atlas*)

$$\varphi_\alpha : U_\alpha \to V_\alpha$$

where $U_\alpha \subseteq M$, $V_\alpha \subseteq \mathbb{R}^d$ are open, φ_α is a homeomorphism, such that

1. $M = \bigcup_\alpha U_\alpha$. That is, the charts cover M.
2. Given any charts $\varphi_\alpha, \varphi_\beta$, the map

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$$

   is a smooth map between open subsets of $\mathbb{R}^d$.
3. If φ is *compatible* (as in the previous point) with every chart in the atlas, then φ is a chart in the atlas.

In practice (i.e. using Zorn's lemma), given any collection of charts satisfying the first two conditions, we can find a maximal atlas containing it.

Remark 1.2.
1. The second condition implies that $\varphi_\beta \circ \varphi_\alpha^{-1}$ is a diffeomorphism.
2. d is fixed, assuming that M is connected.

Definition 1.3 (manifold)
A *(smooth) manifold* M is a second countable Huasdorff space topological space, equipped with a smooth structure.
$d = \dim(M)$ is the *dimension* of M.

In practice, we may induce a topology on M from a smooth structure (without the mention of open or homeomorphism), then $D \subseteq M$ is open if for any chart φ_α, $\varphi_\alpha(D \cap U_\alpha)$ is an open subseteq of $\mathbb{R}^d$.

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Remark 1.4. 1. We can also define a C^k manifold by replacing smooth with C^k as above. If $k = 0$, then we have a *topological manifold*.

2. On the other hand, we can replace $\mathbb{R}^d$ with $\mathbb{C}^n$, and obtain a *complex manifold*.

$\mathbb{R}^n$ is (trivially) a manifold.

Example 1.5 (unit sphere)

Define

$$S^n = \{x = (x_0, \dots, x_n) \in \mathbb{R}^n \mid ||x|| = 1\} \subseteq \mathbb{R}^{n+1}$$

For the charts, we can consider the stereographic projections

$$\varphi(x) = \frac{1}{1 - x_0}(x_1, \dots, x_n) \qquad \text{for } x_0 \neq 1$$

and

$$\psi(x) = \frac{1}{1 + x_0}(x_1, \dots, x_n) \qquad \text{for } x_0 \neq -1$$

In particular,

$$\psi \circ \varphi^{-1}(u) = \frac{u}{||u||^2}$$

Example 1.6

If M, N are manifolds of dimensions d, e respectively, then $M \times N$ is a manifold of dimension $d + e$.

As a corollary, $T^n = (S^1)^n$ is a manifold.

Example 1.7

If M is a manifold, $X \subseteq M$ is open, then X is a manifold.

Lecture 2

A convention, we will will assume for any manifold M, we will assume all components are of the same dimension. In particular, $\dim(M)$ is well defined.

Notation 1.8. We will write M^d to say M is a manifold of dimension d.

Example 1.9 (real projective space)

The real projective space is

$$\mathbb{RP}^n = \{\text{straight lines in } \mathbb{R}^{n+1} \text{ through } 0\}$$

Points in $\mathbb{RP}^n$ can be written using homogeneous coordinates $(x_0 : \cdots : x_n)$, where the x_i are not all zero, and

$$(x_0 : \cdots : x_n) = (\lambda x_0 : \cdots : \lambda x_n)$$

for all $\lambda \neq 0$. We will induce a topology from a smooth structure. The charts are given by (φ_i, U_i) where

$$U_i = \{x_i \neq 0\}$$

and we define

$$\varphi_i(x_0 : \cdots : x_n) = \left(\frac{x_0}{x_i}, \dots, \frac{x_{i-1}}{x_i}, \frac{x_{i+1}}{x_i}, \dots, \frac{x_n}{x_i}\right) \in \mathbb{R}^n$$

For $i < j$,

$$\varphi_j \circ \varphi_i^{-1}(y_0, \dots, y_n) = \left(\frac{y_1}{y_j}, \dots, \frac{1}{y_j}, \dots, \frac{y_n}{y_j} \right)$$

where the 1 is in the i-th position, and we omit the j-th position. Thus, $\mathbb{RP}^n$ is an n-dimensional manifold. Similarly, we can define the complex projective space $\mathbb{CP}^n$, which is a $2n$-manifold (and a n-dimensional complex manifold).

Example 1.10 (Grassmannian)

$$\mathrm{Gr}(k, n) = \{k\text{-dimensional subspaces of } \mathbb{R}^n\}$$

Then $\mathrm{Gr}(k, n)$ is a manifold of dimension $k(n-k)$. Note we can also define the Grassmannian over $\mathbb{C}$.

Sketch proof. An example of a coordinate neighbourhood is

$$U = \left\{k\text{-dimensional subspaces obtainable as the span of rows of } k \times n \text{ matrices of the form } \begin{pmatrix} I_k & * \end{pmatrix}\right\}$$

where I_k is the $k \times k$ identity matrix. In this case, the $k \times (n-k)$ block $*$ defines local coordinates. Choosing different columns, we have $\binom{n}{k}$ neighbourhoods, $U_{i_1,\dots,i_k}$, where $i_1 < i_2 < \cdots < i_k$. □

In particular, note that $\mathbb{RP}^n = \mathrm{Gr}(1, n+1)$.

Example 1.11 (Non-example)

Consider an equivalence relation on $\mathbb{R}^2$ given by

$$(x, y) \sim \left(\lambda x, \frac{1}{\lambda} y \right)$$

for any $\lambda \neq 0$. Let $X = \mathbb{R}^2/\sim$ for the quotient, with the quotient topology. We can consider

$$\{(x, y) \mid xy = c\}$$

If $c \neq 0$, then this is one equivalence class. If $c = 0$, this splits into three, which are

- $\{(0, 0)\}$,
- $\{(x, 0) \mid x \neq 0\}$,
- $\{(0, y) \mid y \neq 0\}$

We can write

$$X = (-\infty, 0) \cup (0, \infty) \cup \{0', 0'', 0'''\}$$

We have three charts φ_i corresponding to each choice of zero. For example,

$$U_1 = (-\infty, 0) \cup \{0'\} \cup (0, \infty)$$

with the "natural" map φ_1. On the other hand, the induced topology is not Hausdorff. Note the induced topology is *not* the quotient topology.

Refer to Examples Sheet 1 Question 12 for an example of a non-second countable (non-)example.

Definition 1.12 (smooth map)

Let M, N be manifolds, a continuous map $f : M \to N$ is *smooth* if for any $p \in M$, we can find charts (φ, U) near p, (ψ, V) near $f(p)$, and the composition

$$\psi \circ f \circ \varphi^{-1}$$

is smooth. as a map on

$$\varphi(U \cap f^{-1}(V))$$

Definition 1.13 (diffeomorphism)

A smooth map $f : M \to N$ is a *diffeomorphism* if f is a bijection, and f^{-1} is smooth. In this case, we say that M, N are *diffeomorphic*.

Proposition 1.14.

1. If $U \subseteq \mathbb{R}^n$, then $f : U \to \mathbb{R}^m$ is smooth in the above definition if and only if it is smooth in the sense of multi-variate calculus.
2. Every chart $\varphi : U \to \varphi(U) \subseteq \mathbb{R}^d$ is a smooth map of manifolds. Moreover, it is a diffeomorphism.
3. If $f : M \to N$, $g : N \to P$ are smooth, then $g \circ f : M \to P$ is smooth.

Proof. Obvious, and so omitted. □

1.1 Matrix Lie groups

Consider the group $\mathrm{GL}(n, \mathbb{R}) \subseteq \mathbb{R}^{n^2}$. In fact, this is an open subset, and so it is a manifold. Moreover, matrix multiplication and inversion are smooth.

Definition 1.15 (Lie group)

A group G is called a *Lie group* if

- G is a manifold,
- the multiplication map $G \times G \to G$ is smooth,
- inversion $G \to G$ is smooth.

Remark 1.16. Suffices to show that the map

$$G \times G \ni (\sigma, \tau) \mapsto \sigma\tau^{-1} \in G$$

is smooth.

In particular, $\mathrm{GL}(n, \mathbb{R}$ is a Lie group. Similarly, $\mathrm{GL}(n, \mathbb{C})$ is also a Lie group. More generally, for $A = (a_{ij}) \in \mathrm{Mat}_n(\mathbb{C})$, we define the norm $\|A\| = n \max_{i,j} |i, j|$. (Note in finite dimensional, all norms are equivalent). In particular,

$$\|AB\| \leq \|A\|\|B\|$$

We define the *exponential of* A as

$$\exp(A) = \sum_k \frac{A^k}{k!}$$

This series converges for all A, by the same proof as for $\exp : \mathbb{C} \to \mathbb{C}$ (or more generally, any Banach algebra). Moreover, it converges uniformly on balls

$$\{\|A\| \leq R\}$$

by the Weierstrass M-test. In particular, $\exp : \mathrm{Mat}_n(\mathbb{C}) \to \mathrm{Mat}_n(\mathbb{C})$ is continuous.

Lecture 3

First of all, note that

$$f(A) = A^n$$

is a differentiable function, with

$$\|(\mathrm{d}f)_A\| \leq n|A|^{n-1}$$

Therefore, the series which we get by differentiating term by term, converges uniformly on any bounded set, again by the Weierstrass M-test.

Therefore, $\exp(A)$ is (at least) C^1 on $\mathrm{Mat}_n(\mathbb{C})$. We can similarly consider the higher derivatives, and we find that $\exp(A)$ is C^∞.

Remark 1.17. Some basic properties of the matrix exponential:

1. $\exp\left(A^{\mathsf{T}}\right) = \exp(A)^{\mathsf{T}}$,
2. $\exp(A^*) = \exp(A^*)$,
3. $\exp\left(CAC^{-1}\right) = C\exp(A)C^{-1}$,

In general,

$$\exp(A+B) = \exp(A)\exp(B)$$

does not hold. However, it does hold if $AB = BA$. In particular,

4. $\exp(A)\exp(-A) = \exp(-A)\exp(A) = I$

We can also define the *matrix logarithm* via the series

$$\log(I+A) = \sum_m (-1)^{m+1}\frac{A^m}{m}$$

We can show that this converges absolutely and uniformly on $\{|A| < \varepsilon\}$ for any $0 < \varepsilon < 1$. The same result holds for the term by term differentiation, and again we get that $\log(I+A)$ is C^∞ on $\{|A| < 1\}$

Suppose $\log(A)$ is defined, that is, $|A - I| < 1$, then

$$\exp(\log(A)) = A$$

since the same proof as in $\mathbb{C}$ works for any Banach algebra.

On the other hand, if we would like

$$\log(\exp(A)) = A$$

we clearly need $|\exp(A) - I| < 1$. Suppose

$$A_\theta = \begin{pmatrix} 0 & -\theta \\ \theta & 0 \end{pmatrix}$$

for some $\theta \in \mathbb{R}$. In particular,

$$A_\theta^2 = -\theta^2 I$$

and we find that

$$\exp(A_\theta) = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$$

Setting $\theta = 2\pi$, then we find that $\exp(A_{2\pi}) = I$, but

$$\log(\exp(A_{2\pi})) = \log(I) = 0 \neq A_{2\pi}$$

Hence $|\exp(A) - I| < 1$ is a necessary, but not sufficient. On the other hand, $|A| < \log(2)$ is sufficient, since in this case,

$$\left|e^{|A|} - 1\right| < 1$$

which is required for absolute convergence.

Example 1.18 (orthogonal group)

Consider

$$O(n) = \left\{A \in \mathrm{GL}(n,\mathbb{R}) \mid AA^{\mathsf{T}} = A^{\mathsf{T}}A = I\right\}$$

We will show that $O(n)$ is a Lie group. Let A be an orthogonal matrix, $|A - I| < \varepsilon < 1$, then $B = \log(A)$ is defined. Moreover, $\exp(B) = A$. By continuity, we can take ε small enough such that $|B| < \log(2)$. Now

$$\exp(B)\exp\left(B^{\mathsf{T}}\right) = AA^{\mathsf{T}} = \mathrm{id}$$

and so

$$\exp(B) = A = (A^{\mathsf{T}})^{-1} = \exp\left(B^{\mathsf{T}}\right)^{-1} = \exp\left(-B^{\mathsf{T}}\right)$$

Taking log, since $|B| = \left|B^{\mathsf{T}}\right| < \log(2)$, we see that $B = -B^{\mathsf{T}}$.

Conversely, if B is skew-symmetric, with $|B| < \log(2)$, then

$$\exp(B)\exp\left(B^{\mathsf{T}}\right) = I$$

Define

$$V_0 = \left\{B \in \mathrm{Mat}_n(\mathbb{R}) \mid |B| < \log(2), B^{\mathsf{T}} + B = 0\right\}$$

and define

$$U = \exp(V_0)$$

which is an open neighbourhood of $I \in O(n)$.

Proposition 1.19. *$O(n)$ has a C^∞ structure, making it into a manifold and a Lie group, with dimension $n(n-1)/2$.*

Proof. We will use the subspace topology induced from $O(n) \subseteq \mathbb{R}^{n^2}$. Define

$$\begin{aligned} h : U &\to V_0 \\ h(A) &= \log(A) \end{aligned}$$

This is a homeomorphism onto its image, which is an open neighbourhood of $0 \in \mathfrak{so}(n) \cong \mathbb{R}^{n(n-1)/2}$.

Given $C \in O(n)$, define

$$U_C = \{CA \mid A \in U\}$$

and

$$\begin{aligned} h_C : U_C &\to V_0 \\ h_C(A) &= \log\left(C^{-1}A\right) \end{aligned}$$

Again, this is a homeomorphism onto its image. Since $C \in U_C$, we have an open cover. Moreover,

$$h_{C_2} \circ h_{C_1}^{-1}(B) = h_{C_2}(C_1\exp(B)) = \log\left(C_2^{-1}C_2\exp(B)\right)$$

which is a smooth map between open neighbourhoods of $\mathfrak{so}(n)$. Therefore, $O(n)$ is a smooth manifold. Similarly, for consider the map

$$F(A_1, A_2) = A_1A_2^{-1}$$

In local coordinates,

$$\begin{aligned} F_{\mathrm{loc}}(B_1, B_2) &= h_{C_1C_2^{-1}}(F(h_{C_1}^{-1}(B_1), h_{C_2}^{-1}(B_2))) \\ &= \log\left((C_1C_2^{-1})^{-1}C_1\exp(B_1)(C_2\exp(B_2))^{-1}\right) \\ &= \log\left(C_2\exp(B_1)\exp(-B_2)C_2^{-1}\right) \end{aligned}$$

and we can see that this is a smooth map as a function of B_1, B_2. □

This method works for more groups, and is sometimes called the Cayley construction (for matrix Lie groups).

2 Tangent spaces to manifolds

Consider a local curve in a ball in $\mathbb{R}^n$, defined by a smooth parametrisation $x(t) = (x_1(t), \ldots, x_n(t))$. Suppose we have that $x(0) = p \in \mathbb{R}^n$. In this case, its tangent vector at p is

$$\dot{x}(0) \in \mathbb{R}^n$$

If $y(x)$ is a smooth change of coordinates near p, then by the chain rule, we have

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} y(x(t)) = J(y)_p\dot{x}(0)$$

where $J(y)$ is the Jacobian matrix. More explicitly,

$$\dot{y}_i(0) = \sum_j \frac{\partial y_j}{\partial x_j} \dot{x}_j(0)$$

Definition 2.1 (tangent vector)

A *tangent vector* a to a manifold M^n at a point $p \in M$, is an assignment to each coordinate chart (U, φ) with $p \in U$, an n-tuple of coordinates $(a_1, \ldots, a_n) \in \mathbb{R}^n$, such that if (U', φ') is another chart near p, $(x_i), (x_i')$ the respective local coordinates,

$$a_i' = \sum_j \frac{\partial x_i'}{\partial x_j} a_j$$

This definition is called the *tensorial definition*, where we focus on the transformation law. There are several other definitions, for example, derivations, equivalence of curves to first order and so on.

Definition 2.2 (tangent space)

The *tangent space* to M at a point p, denoted T_pM is the space of all tangent vectors to M at p. It is naturally a vector space (of dimension n over $\mathbb{R}$).

Lecture 4

A choice of chart (with local coordinates (x_i) say), determines a linear isomorphism

$$\mathrm{T}_pM \to \mathbb{R}^n$$

where $n = \dim(M)$. We denote by $\frac{\partial}{\partial x_i}$, the basis corrsponding to the usual standard basis of $\mathbb{R}^n$, under this linear isomorphism.

If we want to emphasise the point $p \in M$, we may write

$$\left(\frac{\partial}{\partial x_i}\right)_p \quad \text{or} \quad \left.\frac{\partial}{\partial x_i}\right|_p$$

Then by the usual linear algebra, we have that

$$\left(\frac{\partial}{\partial x_i}\right)_p = \sum_j \left(\frac{\partial x_j'}{\partial x_i}\right)_p \left(\frac{\partial}{\partial x_j'}\right)_p$$

An alternative view on tangent vectors is as *derivations*. Given

$$a = \sum a_i \frac{\partial}{\partial x_i} \in \mathrm{T}_pM$$

we can define a *first-order derivation* at p,

$$a : C^\infty(M) \to \mathbb{R}$$

where

$$a(f) = \sum_i a_i \frac{\partial f}{\partial x_i}$$

in local coordinates. This definition is independent of the choice of local coordinates. We can define this in a coordinate independent way, by defining

$$a(f) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} f(x(t))$$

where x is a local curve on M, with $\dot{x}(0) = a$. Moreover, every derivation satisfies the *Leibniz rule*

$$a(fg) = f \cdot a(g) + a(f) \cdot g$$

Conversely, every linear map $a : C^\infty(M) \to \mathbb{R}$ satisfying the Leibniz rule above arises as a derivation corresponding to a tangent vector.

Example 2.3

Let $r : \mathbb{R}^2_{u,v} \supseteq D \to S \subseteq \mathbb{R}^3$ be a regular parametrisation for a surface S. Then $\varphi = r^{-1}$ defines a chart. The partial derivatives r_u, r_v correspond to

$$\frac{\partial}{\partial u}, \frac{\partial}{\partial v}$$

respectively.

For a Lie group G, the tangent spaces has an "infinitessimal counterpart" of the group structure.

Definition 2.4 (Lie algebra)

A *Lie algebra* is a vector space $\mathfrak{g}$, wiht a bilinear map

$$[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$$

called the *Lie bracket*, with

- anticommutativity $[x, y] = [-y, x]$,
- the Jacobi idenity

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$$

Theorem 2.5. Let $G \subseteq \mathrm{GL}(n, F)$ be a matrix Lie group (where $F = \mathbb{R}$ or $\mathbb{C}$), such that log defines a chart near $I \in G$. Let

$$\mathfrak{g} = \mathrm{T}_I G$$

is identified with a (real) subspace of $\mathrm{Mat}_n(F)$. Then $\mathfrak{g}$ is a Lie algebra, with

$$[x, y] = xy - yx$$

We call $\mathfrak{g}$ the *Lie algebra of* G. We may also write

$$\mathfrak{g} = \mathrm{Lie}(G)$$

Proof. Clearly $\mathfrak{g}$ is a vector space, and $[\cdot, \cdot]$ is anticommutative and satisfies the Jacobi identity. Therefore, all we need to show is that if $x, y \in \mathfrak{g}$, then so is $[x, y]$.

For $B_1, B_2 \in \mathfrak{g}$, consider

$$A(t) = \exp(B_1 t) \exp(B_2 t) \exp(-B_1 t) \exp(-B_2 t)$$

This defines a smooth path in G, with $A(0) = I$. Recall

$$\exp(Bt) = I + Bt + \frac{1}{2} B^2 t^2 + o(t^2)$$

as $t \to 0$. We obtain that

$$A(t) = I + [B_1, B_2] t^2 + o(t^2)$$

Letting $B(t) = \log(A(t))$, for small t, we have that $B(t) = [B_1, B_2] t^2 + o(t^2)$. But we also have that

$$\exp(A(t)) = B(t)$$

Moreover, by definition, $B(t) \in \mathfrak{g}$ for t small, as $B(t)$ is in the image of the logarithm chart. So

$$\frac{B(t)}{t^2} \in \mathfrak{g}$$

as $\mathfrak{g}$ is a vector space. So

$$\lim_{t \to 0} \frac{B(t)}{t^2} = [B_1, B_2] \in \mathfrak{g}$$

since $\mathfrak{g} \leq \mathrm{Mat}_n(F)$ is closed. □

If we take $G = O(n)$, then we have that

$$\mathfrak{g} = \mathfrak{o}(n) = \left\{A \in \mathrm{Mat}_n(\mathbb{R}) \mid A^{\mathsf{T}} + A = 0\right\}$$

is the space of skew-symmetric matrices.

Definition 2.6

Let M be a manifold. Then the *tangent bundle of* M is

$$\mathrm{T}M = \bigsqcup_{p \in M} \mathrm{T}_p M$$

Theorem 2.7. $\mathrm{T}M$ has a smooth structure, making it into a manifold of dimension $2\dim(M)$.

Proof. The topology on $\mathrm{T}M$ will be induced from the smooth structure. Let (φ, U) be a chart on M. Say the local coordinates corresponding to φ are $x_1, \ldots, x_n$. So we can write $a \in \mathrm{T}_p M$ as

$$\sum a_i \frac{\partial}{\partial x_i}$$

Define

$$U_T = \bigsqcup_{p \in U} \mathrm{T}_p M$$
$$\varphi_T(p, a) = (\varphi(p), (a_i))$$

It is clear that the sets of the form U_T cover, so all we need to check is that the change of coordinate maps are smooth. Let (ψ, V) be another chart on M, with local coordinates y_i. Then

$$\psi \circ \varphi^{-1}(x, a) = (y, b)$$

and it is easy to see that $y = \psi \circ \varphi^{-1}(x)$, and

$$b_i = \sum_j \frac{\partial y_i}{\partial x_j} a_j$$

Hausdorff and second countable follow from the fact that M and $\mathbb{R}^n$ are. □

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